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System and method for generating work plan for autonomous operation of compactor

Abstract

A system for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine includes a first controller that receives information pertaining to a work area on which the earthmoving machine is required to perform at least one operation. The system also includes a central controller that receives, from the first controller, information pertaining to the work area on which the earthmoving machine is required to perform the at least one operation and analyzes the work area for virtually segmenting the work area into a plurality of virtual work areas. The central controller also receives, from the first controller, data indicative of a movement of the earthmoving machine through each virtual work area from the plurality of virtual work areas and determines an optimal direction of movement for the compactor based on the data indicative of the movement of the earthmoving machine.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATION**

(1) This application claims priority under 35 USC § 6119 and the Paris Convention to Indian Patent Application 202111058477 filed on Dec. 15, 2021.

TECHNICAL FIELD

(2) The present disclosure relates to a system and a method for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine.

BACKGROUND

(3) Various operations, such as material removal, material movement, and/or material distribution may have to be performed at a worksite before initiating construction of infrastructures such as, for example, roadways, speedways for motor vehicles, airport runways, dams, water spill ways, and residential or commercial facilities.

(4) Typically, worksite preparation may require the use of several types of machines. In some cases, a dozer may be required to perform a grading operation at a particular area of the worksite. Subsequently, the dozed worksite may be followed up with a compaction operation that may require use of a compactor thereon. With advancement of technology, machines are being designed to operate in an autonomous manner as they offer various advantages. For example, autonomous machines may be used to replace involvement or need for human workers/operators especially in unsuitable environments, such as, at high altitudes, or at sparsely populated regions. Further, autonomous machines may operate for longer periods of time as compared to manned machines, thereby increasing productivity. Furthermore, autonomous machines may be operated in accordance with precise control strategies aimed at fulfilling one or more end objectives, for instance, optimizing efficiency while reducing downtime and emissions.

(5) U.S. Pat. No. 6,112,143 discloses an apparatus and method for establishing the perimeter of a work site for a mobile machine capable of operating autonomously or semi-autonomously. The apparatus and method includes activating a learning mode, positioning the mobile machine at a plurality of locations on the perimeter, and recording position data at each location from a sensor system, such as GPS, that provide signals corresponding to the position of the mobile machine. The position data is stored and a data processor associated with a control system generates a display of the perimeter so that an operator may visually verify the size, location, and shape of the work area bounded by the perimeter.

SUMMARY

(6) In one aspect of the present disclosure, a system for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine is provided. The system includes

a first controller associated with the earthmoving machine. The first controller receives information pertaining to a work area on which the earthmoving machine is required to perform at least one operation. The system also includes a central controller coupled in communication with the first controller. The central controller is configured to receive, from the first controller, information pertaining to the work area on which the earthmoving machine is required to perform the at least one operation. The central controller is also configured to analyze the work area for virtually segmenting the work area into a plurality of virtual work areas based on a receipt of the information pertaining to the work area. The central controller is further configured to receive, from the first controller, data indicative of a movement of the earthmoving machine through each virtual work area from the plurality of virtual work areas. The central controller is configured to determine an optimal direction of movement for the compactor on at least one virtual work area from the plurality of virtual work areas based on the data indicative of the movement of the earthmoving machine, such that the compactor is configured to move along the optimal direction of movement during the autonomous operation of the compactor. The central controller is also configured to generate a virtual fence around the at least one virtual work area from the plurality of virtual work areas based on generation of the optimal direction of movement for the compactor. The central controller is further configured to transmit information pertaining to the optimal direction of movement to the compactor on the at least one virtual work area from the plurality of virtual work areas when the earthmoving machine is outside the virtual fence.

(7) In another aspect of the present disclosure, a method for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine is provided. The method includes receiving, by a first controller associated with the earthmoving machine, information pertaining to a work area on which the earthmoving machine is required to perform at least one operation. The method also includes receiving, by a central controller, information pertaining to the work area on which the earthmoving machine is required to perform the at least one operation from the first controller. The central controller is coupled in communication with the first controller. The method further includes analyzing, by the central controller, the work area for virtually segmenting the work area into a plurality of virtual work areas based on a receipt of the information pertaining to the work area. The method includes receiving, by the central controller, data indicative of a movement of the earthmoving machine through each of the plurality of virtual work areas from the first controller. The method also includes determining, by the central controller, an optimal direction of movement for the compactor on at least one virtual work area from the plurality of virtual work areas based on the data indicative of the movement of the earthmoving machine, such that the compactor is configured to move along the optimal direction of movement during the autonomous operation of the compactor. The method also includes generating, by the central controller, a virtual fence around the at least one virtual work area from the plurality of virtual work areas based on generation of the optimal direction of movement for the compactor. The method further includes transmitting, by the central controller, information pertaining to the optimal direction of movement to the compactor on the at least one virtual work area from the plurality of virtual work areas when the earthmoving machine is outside the virtual fence.

(8) In yet another aspect of the present disclosure, a computer readable medium having computer executable instructions for performing a method for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine. The method includes receiving information pertaining to a work area on which the earthmoving machine is required to perform at least one operation. The method also includes analyzing the work area for virtually segmenting the work area into a plurality of virtual work areas based on a receipt of the information pertaining to the work area. The method further includes receiving data indicative of a movement of the earthmoving machine through each of the plurality of virtual work areas. The method includes determining an optimal direction of movement for the compactor on at least one virtual work area from the plurality of virtual work areas based on the data indicative of the movement of the

earthmoving machine, such that the compactor is configured to move along the optimal direction of movement during the autonomous operation of the compactor. The method also includes generating a virtual fence around the at least one virtual work area from the plurality of virtual work areas based on generation of the optimal direction of movement for the compactor. The method further includes transmitting information pertaining to the optimal direction of movement to the compactor on the at least one virtual work area from the plurality of virtual work areas when the earthmoving machine is outside the virtual fence.

(9) Other features and aspects of this disclosure will be apparent from the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a pictorial view depicting an earthmoving machine and a compactor operating at a worksite, according to an exemplary embodiment of the present disclosure;
- (2) FIG. 2 is a block diagram depicting a system for generating a work plan for autonomous operation of the compactor in tandem with the earthmoving machine, according to an exemplary embodiment of the present disclosure;
- (3) FIG. 3 is a schematic view of an exemplary work area on which the earthmoving machine may operate;
- (4) FIG. 4A is a schematic view illustrating various structured virtual work areas associated with another exemplary work area;
- (5) FIG. 4B is a schematic view illustrating an exemplary unstructured virtual work area associated with yet another exemplary work area;
- (6) FIG. 5 is a schematic view illustrating various exemplary movement patterns associated with the earthmoving machine and an exemplary optimal direction of movement for the compactor;
- (7) FIG. 6 is a schematic view of an exemplary virtual work area including a continuous edge;
- (8) FIG. 7 is a schematic view illustrating exemplary data received from a first controller associated with the earthmoving machine;
- (9) FIG. 8 is a schematic view of an exemplary virtual work area on which the earthmoving machine may operate;
- (10) FIG. 9 is a flowchart depicting a method for generating the work plan for the autonomous operation of the compactor in tandem with the earthmoving machine, according to an exemplary embodiment of the present disclosure; and
- (11) FIG. 10 is a process flowchart for low-level implementation of the method of FIG. 9, according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

(12) Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. Wherever possible, corresponding or similar reference numbers will be used throughout the drawings to refer to the same or corresponding parts.

(13) Referring to FIG. 1, a portion of an exemplary worksite **100** is illustrated. The worksite **100** may include, for example, a mine site, a land fill, a quarry, and a construction site. The worksite **100** may be any worksite, such as, for construction of roadways or access roads, residential complexes, or commercial complexes, or any other type of worksite on which work or labor may be performed.

(14) In one example, the worksite **100** may undergo geographic alterations due to work being performed thereon by one or more earthmoving machines. For example, an earthmoving machine **102** embodied as a dozer may operate at the worksite **100**. The earthmoving machine **102** includes an implement **104** and a pair of ground engaging members **106** (only one of which is illustrated

herein) embodied as tracks. The earthmoving machine **102** may perform one or more tasks such as a grading operation or any other geography-altering tasks at the worksite **100**. Alternatively, the earthmoving machine **102** may be embodied as an excavator, a tractor, a loader, a grader, a scraper, or any other machine that may perform earthmoving operations.

(15) Further, a compactor **108** may also operate at the worksite **100** and in tandem with the earthmoving machine **102**. For example, the compactor **108** may perform a compaction operation at the worksite **100** previously worked upon by the earthmoving machine **102**. The compactor **108** may be embodied as a soil compactor. In another example, the compactor **108** may be embodied as a pneumatic compactor. The compactor **108** includes a pair of drums **110**. Alternatively, the compactor **108** may include a single drum and a pair of wheels. The compactor **108** disclosed herein is embodied as an autonomous compactor or a semi-autonomous compactor.

(16) For exemplary purposes, one earthmoving machine **102** and one compactor **108** are illustrated in FIG. 1. However, more than one earthmoving machine **102** and more than one compactor **108** may operate at the worksite **100** depending on specific requirements of an application. Further, the earthmoving machine **102** and the compactor **108** may operate on one or more exemplary work areas **300**, **400**, and **406** (shown in FIGS. 3, 4A, and 4B, respectively) defined at the worksite **100**.

(17) Referring to FIG. 2, a system **200** for generating a work plan for autonomous operation of the compactor **108** (see FIG. 1) in tandem with the earthmoving machine **102** (see FIG. 1) is illustrated. The work plan may include details, such as, an optimal direction of movement for the compactor **108**, an entry point **302**, **304** (shown in FIG. 3) from which the compactor **108** may enter the work area **300**, **400**, and **406** (see FIGS. 3, 4A, and 4B, respectively) to be worked on, an exit point **302**, **304** from which the compactor **108** may exit the work area **300**, **400**, and **406** to be worked on, and portions of the work areas **300**, **400**, and **406** on which the compactor **108** is required to perform the compaction operation. For explanatory purposes, the system **200** will now be explained in relation to operations being performed on the work area **300** by the earthmoving machine **102**. However, the description provided below is equally applicable to the other work areas **400**, **406**.

(18) As shown in FIG. 2, the system **200** includes a first controller **202** associated with the earthmoving machine **102** (see FIG. 1). The first controller **202** may be present onboard the earthmoving machine **102**. The first controller **202** receives information pertaining to the work area **300** on which the earthmoving machine **102** is required to perform one or more operations. The information pertaining to the work area **300** may be uploaded to the first controller **202** by an operator or a personnel in charge of the worksite **100** (see FIG. 1) or the earthmoving machine **102**. In an example, the information pertaining to the work area **300** may be prestored within a memory (not shown) associated with the first controller **202**. The one or more operations may include, for example, a grading operation, a material excavation operation, backfill or reclamation, and a material movement operation.

(19) In an example, the information pertaining to the work area **300** may include a virtual design boundary of the work area **300** on which the earthmoving machine **102** is required to perform the one or more operations. The virtual design boundary may include information such as a layout of the work area **300** and a surface area defined by the work area **300**. It should be noted that the virtual design boundary of the work area **300** may be of any arbitrary shape and size.

(20) In an example, in addition to the virtual design boundary of the work area **300**, the first controller **202** may also receive details such as a location of the work area **300** or other parameters such as the presence of buildings, trees, or other objects/obstacles present on the work area **300**. Further, the virtual design boundary of the work area **300** may be displayed on a display device (not shown) present within the earthmoving machine **102**. Based on the receipt of the information pertaining to the work area **300**, an operator of the earthmoving machine **102** may initiate operation of the earthmoving machine **102** for executing the one or more operations to be performed on the work area **300**. In an example, when the earthmoving machine **102** initiates the one or more operations on the work area **300**, the first controller **202** may transmit an elevation data of the work

area **300** to a central controller **204**.

(21) Further, the system **200** also includes the central controller **204** coupled in communication with the first controller **202**. The central controller **204** may execute instructions stored on a computer readable medium to perform methods for generating the work plan for the autonomous operation of the compactor **108** in tandem with the earthmoving machine **102**. The central controller **204** may be located at a back office or a remote operator station. The central controller **204** may be in communication with a memory **206** via wired means or wireless means. The wireless means may include, for example, Wi-Fi, Bluetooth, or any other such means. The memory **206** may include, for example, a flash memory, a random-access memory (RAM), and an electrically erasable programmable read-only memory (EEPROM). The memory **206** may store data, such as, algorithms, instructions, and arithmetic operations. The central controller **204** may execute various types of digitally stored instructions such as a software or an algorithm, retrieved from the memory **206**, or a firmware program which may enable the central controller **204** to perform a wide variety of operations. Although aspects of the present disclosure may be described generally as being stored in the memory **206**, it may be contemplated that these aspects can be stored on or read from different types of computer program products or computer-readable media, such as, computer chips and secondary storage devices including hard disks, floppy disks, optical media, compact disc-read only memory (CD-ROM), or other forms of RAM or read only memory (ROM). In some examples, the work plan generated by the central controller **204** may be stored in the memory **206**.

(22) The central controller **204** receives the information pertaining to the work area **300** on which the earthmoving machine **102** is required to perform the one or more operations from the first controller **202**. For example, the central controller **204** may receive the elevation data and/or the virtual design boundary of the work area **300** from the first controller **202**. Moreover, the central controller **204** may receive data corresponding to the one or more operations being performed by the earthmoving machine **102**. The information may be transmitted via wireless means, such as, Wi-Fi, Bluetooth, or any other such means. In an example, the data received from the first controller **202** may be stored in the memory **206** of the central controller **204** for further processing.

(23) As shown in FIG. 3, the central controller **204** (see FIG. 2) analyzes the work area **300** for virtually segmenting the work area **300** into a number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on a receipt of the information pertaining to the work area **300**. More particularly, the central controller **204** may analyze one or more convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** and one or more concave vertices **316A**, **316B**, **316C**, **316D**, **316E** of the work area **300** for virtually segmenting the work area **300** into the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. The terms “convex vertex” as used herein relates to a vertex on the virtual design boundary having an angle that is less than 180 degrees. The terms “concave vertex” as used herein relates to a vertex on the virtual design boundary having an angle that is greater than 180 degrees.

(24) The work area **300** may be virtually segmented based on factors, such as, a length of each virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** and/or the surface area of the work area **300**, without any limitation thereto. In an example, a length of each virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** may lie between 20 foot and 70 foot on ground, without any limitation thereto. In an example, a width of each virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** may lie between 5 foot and 30 foot on ground, without any limitation thereto. It should be noted that although the work area **300** is virtually segmented along a length of the work area **300**, it may be contemplated that the work area **300** may be virtually segmented along a width of the work area **300**.

(25) The central controller **204** may determine one or more entry points **302** for the compactor **108** and exit points **304** for the compactor **108** based on an analysis of the number of convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I**. As illustrated, the work area **300**

includes nine convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I**. It should be noted that the entry point **302** and the exit point **304** may be determined based on an analysis of a direction of movement of the earthmoving machine **102** (see FIG. 1) on the work area **300**. Accordingly, in an example, the entry point **302** may be defined at any of a first end **308**, a second end **310**, or a third end **312** based on the direction of movement that the earthmoving machine **102** may have previously followed, without any limitation thereto. Moreover, in an example, the exit point **304** may be defined at any of the first end **308**, the second end **310**, or the third end **312** based on the direction of movement that the earthmoving machine **102** may have previously followed, without any limitation thereto.

(26) For determining the entry point **302** and the exit point **304**, the central controller **204** may determine all the convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** present on the virtual design boundary corresponding to the work area **300**. If an effective angle change on edges with the convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** is equal to 180 degrees, the central controller **204** may assign two of those convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** as the entry point **302** and two of those convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** as the exit point **304**. In an example, the central controller **204** may determine an angle **P1** defined by each convex vertex **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I**. Further, the central controller **204** may determine from the virtual design boundary if any two convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** are located in succession to each other along a periphery of the virtual design boundary. Moreover, if a summation of the angles **P1** defined by any two successively disposed convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** is equal to 180 degrees, the central controller **204** may assign the entry point **302** and/or the exit point **304**.

(27) FIG. 3 illustrates two adjacent convex vertices **314A**, **314B** defined proximate to the first end **308**, two adjacent convex vertices **314D**, **314E** defined proximate to the second end **310**, and two adjacent convex vertices **314F**, **314G** defined proximate to the third end **312**. Further, the summation of the angles **P1** defined by the two convex vertices **314A**, **314B** is substantially equal to 180 degrees. Accordingly, the central controller **204** may assign the entry point **302** and/or the exit point **304** proximate to the first end **308**.

(28) Further, the summation of the angles **P1** defined by the two convex vertices **314D**, **314E** is substantially equal to 180 degrees, the central controller **204** may assign the entry point **302** and/or the exit point **304** proximate to the second end **310**. Similarly, the summation of the angles **P1** defined by the two convex vertices **314F**, **314G** is substantially equal to 180 degrees, the central controller **204** may assign the entry point **302** and/or the exit point **304** proximate to the third end **312**. In an example, the central controller **204** may also generate entry and exit points for each virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**.

(29) Further, the central controller **204** may split the work area **300** into the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on an angle **P2** defined by the number of concave vertices **316A**, **316B**, **316C**, **316D**, **316E**. The number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** may include one or more of a structured virtual work area and an unstructured virtual work area. The term “structured virtual work area” as used herein relates to the virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** that may have a definite shape, such as, a rectangular shape, a square shape, a triangular shape, and a trapezoidal shape, or any other such regular polygonal shape. Further, the term “unstructured virtual work area” as described herein relates to virtual work areas generated on the work area **300** that may have an arbitrary and/or non-definite shapes. It should be noted that the present disclosure is not limited by scope in the determination of a structured/unstructured work area vis-à-vis a shape of the virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**.

(30) The central controller **204** may determine whether a particular virtual work area **306A**, **306B**,

306C, 306D, 306E, 306F, 306G, 306H is a structured work area or an unstructured work area based on the angle **P2** defined at the number of concave vertices **316A, 316B, 316C, 316D, 316E**. Further, the work area **300** includes five concave vertices **316A, 316B, 316C, 316D, 316E**. In an example, the concave vertices **316A, 316B, 316C, 316D, 316E** may be indicative of a bend or a turn on the work area **300** that the earthmoving machine **102** or the compactor **108** may operate on. (31) In an example, when the angle **P2** at each of the concave vertices **316A, 316B** lies approximately between 60 degrees and 140 degrees, the work area **300** may be virtually segmented into the virtual work areas **306A, 306B**. Specifically, each virtual work area **306A, 306B** is divided into a structured virtual work area having a substantially trapezoidal shape. Further, the work area **300** may be virtually segmented to form the virtual work area **306A** based on a joining of the concave vertex **316A** and the convex vertex **314I**. Moreover, the work area **300** may be virtually segmented to form the virtual work area **306B** based on a joining of the concave vertex **316B** and the convex vertex **314H**. Further, as the angle **P2** at the concave vertex **316C** lies approximately between 60 degrees and 140 degrees, the work area **300** may be virtually segmented into the virtual work area **306C**. Specifically, the virtual work area **306C** is divided into a structured virtual work area that is substantially parallelogram shaped. Further, the work area **300** may be virtually segmented to form the virtual work area **306C** based on a joining of the concave vertex **316C** and the convex vertex **314C**. Further, a portion of the work area **300** defined along an edge **318** of the work area **300** may be divided into the three virtual work areas **306D, 306E, 306F** based on a size of the work area **300** defined along the edge **318**. Each virtual work area **306D, 306E, 306F** includes a rectangular shape.

(32) Furthermore, the angle **P2** at the concave vertices **316D, 316E** is approximately equal to 90 degrees. In such an example, the work area **300** may be virtually segmented into the virtual work areas **306G, 306H**. Further, each of the virtual work areas **306G, 306H** illustrated herein are structured virtual work areas having a substantially rectangular shape. Furthermore, the work area **300** may be virtually segmented to form the virtual work area **306G, 306H** based on a joining of the concave vertex **316D** and the concave vertex **316E**. It should be further noted that a portion of the work area **300** defined along an edge **320** of the work area **300** may be divided into the two virtual work areas **306G, 306H** based on a size of the work area **300** defined along the edge **320**.

(33) FIG. 4A illustrates another exemplary technique for virtually segmenting the exemplary work area **400**. As shown in FIG. 4A, the work area **400** defines a concave vertex **402A**. An angle **P3** defined by the concave vertex **402A** is approximately equal to 90 degrees. Further, the work area **400** may be virtually segmented into the virtual work areas **404A, 404B, 404C**. Furthermore, each virtual work area **404A, 404B** illustrated herein is a structured virtual work area having a substantially rectangular shape. Moreover, the work area **400** may also be virtually segmented into the virtual work area **404C**. The virtual work areas **404C** illustrated herein is a structured virtual work area having a substantially square shape.

(34) Referring now to FIG. 4B, another exemplary technique for virtually segmenting the exemplary work area **406** is illustrated. As shown in FIG. 4B, the work area **406** defines a concave vertex **408A**, such that an angle **P4** defined by the concave vertex **408A** is less than 60 degrees. As the concave vertex **408A** is less than 60 degrees, the work area **406** may be virtually segmented into a virtual work area **410A**. It should be noted that the virtual work area **410A** includes an unstructured work area. The unstructured work area may have any arbitrary or non-definite shape.

(35) Referring again to FIGS. 1 and 2, the central controller **204** may also determine if the earthmoving machine **102** has concluded the one or more operations on the one or more virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** (see FIG. 3). The central controller **204** may determine that the one or more operations have been concluded based on information received from the first controller **202**. For example, the operator of the earthmoving machine **102** may provide an input to the first controller **202** that may be indicative of the conclusion of the one or more operations on the one or more virtual work areas **306A, 306B, 306C, 306D, 306E, 306F,**

306G, 306H. The first controller **202** may in turn transmit this information to the central controller **204** via wireless means, such as, Wi-Fi, Bluetooth, or any other such means.

(36) Further, in an example, the central controller **204** may transmit the information indicative of the conclusion of the one or more operations at each of the virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** (see FIG. 3) to a second controller **208** associated with the compactor **108**. The second controller **208** may be coupled in communication with the central controller **204**.

(37) In an example, the central controller **204** may determine the optimal direction of movement on the one or more virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** from the number of virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** based on a conclusion of the one or more operations by the earthmoving machine **102**. Further, the central controller **204** determines the optimal direction of movement for the compactor **108** based on the data indicative of the movement of the earthmoving machine **102**. The compactor **108** moves along the optimal direction of movement during the autonomous operation of the compactor **108**.

Specifically, during autonomous operation, the compactor **108** moves along the optimal direction of movement determined before-hand in accordance with the aforementioned embodiment of the present disclosure. For example, as illustrated in FIG. 5, the earthmoving machine **102** (see FIG. 2) may follow a movement pattern **D1** and **D2**, due to which the earthmoving machine **102** may push some material towards one edge and some material towards another edge. Alternatively, the earthmoving machine **102** may move diagonally and follow a movement pattern **D3**, due to which the earthmoving machine **102** may push the material sideways. Further, the earthmoving machine **102** may follow a movement pattern **D4**, due to which the earthmoving machine **102** may push the material towards an edge from which the earthmoving machine **102** may have entered.

(38) It should be noted that the movement patterns **D1, D2, D3, D4** illustrated herein are exemplary in nature, and the earthmoving machine **102** may follow any other movement pattern, without any limitations. The central controller **204** (see FIG. 2) may analyze each of the movement patterns **D1, D2, D3, D4** to determine an exemplary optimal direction of movement **D5** for the compactor **108** (see FIG. 1). The optimal direction of movement **D5** may typically embody a generalized and simplified pattern that may be generated taking into consideration the movement patterns **D1, D2, D3, D4** of the earthmoving machine **102**.

(39) Referring to FIGS. 2 and 3, in addition to the optimal direction of movement, the central controller **204** may also transmit information that may indicate which of the virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** have been worked upon by the earthmoving machine **102** (see FIG. 1). Thus, the work plan may include the information pertaining to the virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** on which the earthmoving machine **102** has already performed the one or more operations. For example, the second controller **208** may receive information that the earthmoving machine **102** has concluded operations at the virtual work areas **306A, 306B, 306C** and the earthmoving machine **102** is yet to conclude operations at the virtual work areas **306D, 306E, 306F, 306G, 306H**. Based on this information, the compactor **108** may only perform the compaction operation on the virtual work areas **306A, 306B, 306C** on which the earthmoving machine **102** has already performed the one or more operations.

(40) Based on receipt of the work plan from the central controller **204**, the second controller **208** located on the compactor **108** may control one or more components of the compactor **108** to move the compactor **108** for performing the compaction operation. For example, the second controller **208** may control the drums **110** (see FIG. 1) of the compactor **108** for movement of the compactor **108** and subsequently to perform the compaction operation.

(41) In an example, the compactor **108** may move through the virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H** in an order in which the earthmoving machine **102** may have performed the one or more operations on the virtual work areas **306A, 306B, 306C, 306D, 306E, 306F, 306G, 306H**. For example, if the earthmoving machine **102** performs the operations on

the virtual work area **306A** and directly moves to the virtual work area **306C**, the compactor **108** may also perform the compaction operation on the virtual work area **306A** and may then directly move to the virtual work area **306C**.

(42) Referring now to FIG. 6, the exemplary virtual work area **600** is illustrated. In an example, the optimal direction of movement for the compactor **108** (see FIG. 1) may be chosen from three possible directions **D6**, **D7**, **D8** of movement of the earthmoving machine **102** (see FIG. 1) on the virtual work area **600**. The central controller **204** may determine the optimal direction of movement for the compactor **108** based on, for example, a longest continuous edge **606** of the virtual work area **600** along which an overall deflection/change in the angle of the longest continuous edge **606** is less than a threshold angle, for example, less than 20 degrees.

(43) The optimal direction of movement for the compactor **108** may be determined based on position data of the earthmoving machine **102** at different instances of time. More particularly, the central controller **204** (see FIG. 2) may receive a number of position inputs associated with the earthmoving machine **102** from the first controller **202** (see FIG. 2). In an example, each position input from the number of position inputs may be indicative of the position of the earthmoving machine **102** on the one or more virtual work areas **600** with respect to time. In an example, the position inputs may be available in terms of 1×1 square feet polygons with timestamps. For example, as illustrated in FIG. 7, polygons **A1**, **A2**, **A3**, **A4** may be representative of position inputs along the direction **D6** of the earthmoving machine **102** (see FIG. 1). Further, polygons **B1**, **B2**, **B3**, **B4** may be representative of position inputs along the direction **D7** of the earthmoving machine **102**. Moreover, polygons **C1**, **C2**, **C3**, **C4** may be representative of position inputs along the direction **D8** of the earthmoving machine **102**.

(44) Further, the central controller **204** (see FIG. 2) may generate a number of vectors based on an analysis of the number of position inputs. In an example, the polygons **A1**, **A2**, **A3**, **A4**, **B1**, **B2**, **B3**, **B4**, **C1**, **C2**, **C3**, **C4** may be arranged in an ascending time sequence to form the vectors. Furthermore, the central controller **204** may join the polygons **A1**, **A2**, **A3**, **A4**, **B1**, **B2**, **B3**, **B4**, **C1**, **C2**, **C3**, **C4** that are close to each other in time for generating the number of vectors. The vectors generated by the central controller **204** thus provide an indication of a magnitude and a direction of the earthmoving machine **102** to the central controller **204**. For example, in the context of the present disclosure, the vectors generated by the central controller **204** may provide an indication of the directions **D6**, **D7**, **D8** of the earthmoving machine **102** on the virtual work area **600**.

(45) Moreover, the central controller **204** may determine the optimal direction of movement for the compactor **108** based on an average dot product of the number of vectors. In an example, the central controller **204** may determine the average dot product of the number of vectors associated with the longest continuous edge **606** (see FIG. 6) of the virtual work area **600** as shown in FIG. 6 to find the optimal direction of movement for the compactor **108**. Accordingly, in an example, the central controller **204** may determine the optimal direction of movement to be similar to the direction **D6** or the direction **D8**.

(46) Further, if the average dot product of the vectors generated using the data of the polygons **A1**, **A2**, **A3**, **A4** is positive, the central controller **204** may determine that the earthmoving machine **102** may be moving along the direction **D6**. However, if the average dot product of the vectors generated using the data of the polygons **A1**, **A2**, **A3**, **A4** is negative, the central controller **204** may determine that the earthmoving machine **102** may be moving along the direction **D8**. It should be noted that, once the central controller **204** may determine the direction of movement of the earthmoving machine **102**, the central controller **204** may determine an entry point through which the compactor **108** may enter the virtual work area **600** and an exit point through which the compactor **108** may exit the virtual work area **600**.

(47) In an example, it may be possible that the direction of movement of the earthmoving machine **102** may not align with the longest continuous edge **606** of the virtual work area **600**. For example, the earthmoving machine **102** may move along the direction **D7**. In such examples, the central

controller **204** may override the direction **D7** of movement of the earthmoving machine **102** and may determine the optimal direction of movement as the direction **D6** or the direction **D8**.

(48) Referring to FIG. **8**, the central controller **204** may generate a virtual fence **810** around the one or more virtual work areas **306E** from the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on the generation of the optimal direction of movement for the compactor **108**. For sake of simplicity, the virtual fence **810** being generated around the virtual work area **306E** associated with the work area **300** will now be explained in detail. However, it should be noted that the central controller **204** may use the technique mentioned below for generating a corresponding virtual fence for each virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306G**, **306H** (see FIG. **3**) associated with the work area **300**, the virtual work areas **404A**, **404B**, **404C** (see FIG. **4A**) associated with the work area **400** (see FIG. **4A**), the virtual work area **410A** (see FIG. **4B**) associated with the work area **406** (see FIG. **4B**), and the virtual work area **600** (see FIG. **6**). In another example, the central controller **204** may generate a virtual fence (similar to the virtual fence **810**) for the entire work area **300**, **400**, and **406** (see FIGS. **3**, **4A**, and **4B**, respectively). In this example, the central controller **204** may transmit the information pertaining to the optimal direction of movement when the earthmoving machine **102** is outside the virtual fence of the corresponding work area **300**, **400**, and **406**.

(49) Further, an area of the virtual fence **810** may be greater than an area of the one or more virtual work areas **306E**. In an example, the central controller **204** may generate the virtual fence **810** after generating the optimal direction of movement for the compactor **108**. As illustrated, the virtual work area **306E** defines an entry point **802** and an exit point **804**. The entry and exit points **802**, **804** are sections through which the compactor **108** may enter and exit the virtual work area **306E**. The virtual work area **306E** also defines a first side edge **806** and a second side edge **808**. Further, the virtual fence **810** defines a fence entry point **812**, such that a forward pathing distance **V1** is defined between the fence entry point **812** of the virtual fence **810** and the entry point **802** of the virtual work area **306E**. It should be noted that the fence entry point **812** of the virtual fence **810** may be indicative of a location from which the compactor **108** may enter the virtual fence **810**. In an example, the forward pathing distance **V1** may lie in a range of 10 meters to 20 meters, without any limitation thereto.

(50) Furthermore, the virtual fence **810** defines a fence exit point **814**, such that a rearward pathing distance **V2** is defined between the fence exit point **814** of the virtual fence **810** and the exit point **804** of the virtual work area **306E**. It should be noted that the fence exit point **814** is a location of the virtual fence **810** from which the compactor **108** may exit the virtual fence **810**. In an example, the rearward pathing distance **V2** may lie in a range of 5 meters to 15 meters, without any limitation thereto.

(51) Moreover, the virtual fence **810** also includes a first fence side edge **816**, such that a first lean out distance **V3** is defined between the first fence side edge **816** of the virtual fence **810** and the first side edge **806** of the virtual work area **306E**. Additionally, the virtual fence **810** includes a second fence side edge **818**, such that a second lean out distance **V4** is defined between the second fence side edge **818** of the virtual fence **810** and the second side edge **808** of the virtual work area **306E**. It should be noted that the term “first lean out distance” and the term “second lean out distance” as used herein may be regarded as an additional distance by which the compactor **108** can move beyond the corresponding edges **806**, **808** of the virtual work area **306E**. In an example, each of the first lean out distance **V3** and the second lean out distance **V4** may lie in a range of 1 meters to 5 meters, without any limitation thereto. The central controller **204** may determine the optimal direction of movement based on the conclusion of the one or more operations by the earthmoving machine **102**. Further, the central controller **204** transmits the information pertaining to the optimal direction of movement to the compactor **108** when the earthmoving machine **102** is outside the virtual fence **810** generated around the one or more virtual work areas **306E**. For example, when the earthmoving machine **102** may move outside the virtual fence **810** and enter an adjacent virtual

work area, the central controller **204** may transmit the optimal direction of movement to the second controller **208**. For a grading operation, the central controller **204** may also determine if an on-grade at the virtual work area **306E** is more than a threshold, for example, 80%. If the on-grade is more than the threshold, the central controller **204** may transmit the work plan containing the information pertaining to the optimal direction of movement to the second controller **208**.

(52) It should be noted that data pertaining to the virtual fence **810** that is generated by the central controller **204** is transmitted to the second controller **208**. If the earthmoving machine **102** concludes the one or more operations on the virtual work area **306E** and exits the corresponding virtual fence **810**, the compactor **108** enters the virtual fence **810** for performing the compaction operation. It should be noted that the second controller **208** is programmed to prevent a movement of the compactor **108** outside the virtual fence **810** until the compaction operations are concluded on the virtual work area **306E**.

(53) It should be noted that, in addition to the optimal direction of movement for the compactor **108**, the central controller **204** may also transmit information pertaining to the virtual work areas **306A**, **306B**, **306C**, **306D**, **306F**, **306G**, **306H** on which the earthmoving machine **102** may have concluded the one or more operations. In an example, the central controller **204** may also transmit information pertaining to the virtual fence **810**, the fence entry point **812** from which the compactor **108** may enter the virtual fence **810**, and/or the fence exit point **814** from which the compactor **108** may exit the virtual fence **810**.

(54) As illustrated in FIG. **1**, the earthmoving machine **102** is operating in the virtual work area **306B** and the compactor **108** is operating in the virtual work area **306A**. When the earthmoving machine **102** concludes the operations in the virtual work area **306B** and moves outside a virtual fence (not shown) of the virtual work area **306B**, the central controller **204** may transmit the work plan to the second controller **208**. Further, once the compactor **108** concludes the compaction operations on the virtual work area **306A**, the compactor **108** may enter the virtual work area **306B** for performing the compaction operations.

(55) It should be noted that each of the first controller **202**, the second controller **208**, and the central controller **204** may embody a single microprocessor or multiple microprocessors for receiving various input signals. Numerous commercially available microprocessors may be configured to perform the functions of respective ones of the first controller **202**, the second controller **208**, and the central controller **204**. Each of the first controller **202**, the second controller **208**, and the central controller **204** may include a central processing unit, a graphics processing unit, and/or any other system associated hardware and networking circuitry, for instance, signal conditioning and/or processing circuitry.

(56) FIG. **9** illustrates a flowchart depicting a method **900** for generating the work plan for the autonomous operation of the compactor **108** in tandem with the earthmoving machine **102**. Referring to FIGS. **1**, **2**, **3**, and **9**, at step **902**, the first controller **202** associated with the earthmoving machine **102** receives the information pertaining to the work area **300** on which the earthmoving machine **102** is required to perform one or more operations. At step **904**, the central controller **204** receives the information pertaining to the work area **300** on which the earthmoving machine **102** is required to perform the one or more operations from the first controller **202**. The central controller **204** is coupled in communication with the first controller **202**.

(57) At step **906**, the central controller **204** analyzes the work area **300** for virtually segmenting the work area **300** into the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on the receipt of the information pertaining to the work area **300**. The central controller **204** analyzes one or more of the number of convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I** and one or more of the number of number of concave vertices **316A**, **316B**, **316C**, **316D**, **316E** of the work area **300** for virtually segmenting the work area **300** into the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. In an example, the central controller **204** virtually segments the work area **300** into the number of virtual work

areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on the angle **P2** defined by the number of concave vertices **316A**, **316B**, **316C**, **316D**, **316E**. The number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** includes one or more of the structured virtual work area and the unstructured virtual work area. Further, the central controller **204** determines one or more of the entry point **302** for the compactor **108** and the exit point **304** for the compactor **108** based on the analysis of the number of convex vertices **314A**, **314B**, **314C**, **314D**, **314E**, **314F**, **314G**, **314H**, **314I**.

(58) At step **908**, the central controller **204** receives the data indicative of the movement of the earthmoving machine **102** through each of the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** from the first controller **202**. At step **910**, the central controller **204** determines the optimal direction of movement for the compactor **108** on the one or more virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** from the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on the data indicative of the movement of the earthmoving machine **102**, such that the compactor **108** moves along the optimal direction of movement during the autonomous operation of the compactor **108**.

(59) Referring now to FIGS. **1**, **2**, **6**, and **9**, in an example, the central controller **204** receives the number of position inputs associated with the earthmoving machine **102** from the first controller **202**. Each position input from the number of position inputs is indicative of the position of the earthmoving machine **102** on the one or more virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** with respect to time. Further, the central controller **204** generates the number of vectors based on the analysis of the number of position inputs. Furthermore, the central controller **204** determines the optimal direction of movement for the compactor **108** based on the average dot product of the number of vectors.

(60) Moreover, referring to FIGS. **1**, **2**, **8**, and **9**, at step **912**, the central controller **204** generates the virtual fence **810** around the one or more virtual work areas **306E** from the number of virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** based on the generation of the optimal direction of movement for the compactor **108**. The central controller **204** generates the virtual fence **810** such that the area of the virtual fence **810** is greater than the area of the one or more virtual work areas **306E**. In an example, the central controller **204** determines the optimal direction of movement based on the conclusion of the one or more operations by the earthmoving machine **102**. Further, at step **914**, the central controller **104** transmits the information pertaining to the optimal direction of movement to the compactor **108** when the earthmoving machine **102** is outside the virtual fence **810** generated around the one or more virtual work areas **306E**. Further, the central controller **204** transmits the information pertaining to the optimal direction of movement to the second controller **208** associated with the compactor **108** when the earthmoving machine **102** is outside the virtual fence **810**. The second controller **208** is coupled in communication with the central controller **204**. It may be desirable to perform one or more of the steps associated with the method **900** in an order different from that depicted. Furthermore, various steps could be performed together.

(61) FIG. **10** is a process flowchart **1000** for low-level implementation of the method **900** of FIG. **9**. Referring to FIGS. **1**, **2**, **3**, and **10**, the process **1000** may be executed by the central controller **204**. The process **1000** may be stored within the memory **206** and may be retrieved for execution by the central controller **204**. At a block **1002**, the process **1000** starts or begin operation. At a block **1004**, the central controller **204** determines if the earthmoving machine **102** has initiated the one or more operations on the work area **300**. At the block **1004**, if the central controller **204** determines that the earthmoving machine **102** has not initiated the one or more operations on the work area **300**, the process **1000** moves to a block **1006**. At the block **1006**, the central controller **204** may wait until the central controller **204** receives information indicative of the initiation of the one or more operations on the work area **300** from the first controller **202**.

(62) However, at the block **1004**, if the central controller **204** determines that the earthmoving

machine **102** has initiated the one or more operations on the work area **300**, the process **1000** moves to a block **1008**. At the block **1008**, the central controller **204** determines if the work area **300** has been virtually segmented. At the block **1008**, if the central controller **204** determines that the work area **300** has not been virtually segmented, the process **1000** moves to a block **1010** to virtual segment the work area **300**.

(63) However, at the block **1008**, if the central controller **204** determines that the work area **300** has been virtually segmented, the process **1000** moves to a block **1012**. At the block **1012**, the central controller **204** determines if the earthmoving machine **102** has concluded the one or more operations on the one or more virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. At the block **1012**, if the central controller **204** determines that the earthmoving machine **102** has not concluded the one or more operations on the one or more virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**, the process **1000** moves to a block **1014**.

(64) At the block **1014**, the central controller **204** may wait until the central controller **204** receives information indicative of the conclusion of the one or more operations on the one or more virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** from the first controller **202**. However, at the block **1014**, if the earthmoving machine **102** has concluded the one or more operations on the one or more virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**, the process **1000** moves to a block **1016**. At the block **1016**, the central controller **204** determines the optimal direction of movement for the compactor **108**.

(65) The process **1000** then moves to a block **1018**. Referring now to FIGS. **1**, **2**, **8**, and **10**, at the block **1018**, the central controller **204** determines if the virtual fence **810** has been generated. At the block **1018**, if the central controller **204** determines that the virtual fence **810** has not been generated, the process **1000** moves to a block **1020**. At the block **1020**, the central controller **204** generates the virtual fence **810**. However, at the block **1018**, if the central controller **204** determines that the virtual fence **810** has been generated, the process **1000** moves to a block **1022**. At the block **1022**, the central controller **204** determines if the earthmoving machine **102** is outside the virtual fence **810**. At the block **1022**, if the central controller **204** determines that the earthmoving machine **102** is not outside the process **1000** moves to a block **1024**. At the block **1024**, the central controller **204** may wait until the earthmoving machine **102** is outside the virtual fence **810**.

(66) However, at the block **1022**, if the central controller **204** determines that the earthmoving machine **102** is outside the virtual fence **810**, the process **1000** moves to a block **1026**. At the block **1026**, the central controller **204** transmits the optimal direction of movement to the second controller **208** associated with the compactor **108**. Further, the process **1000** terminates or ends operation at a block **1028**.

INDUSTRIAL APPLICABILITY

(67) The present disclosure relates to the system **200** and the method **900** for generating the work plan for the autonomous operation of the compactor **108** in tandem with the earthmoving machine **102**. For explanatory purposes, this section will be explained in relation to operations being performed on the work area **300** by the earthmoving machine **102**.

(68) The earthmoving machine **102** may perform the operations at the work areas **300** which may be large in size and may have a complex/irregular shape. However, it may be challenging for the compactor **108** to handle the work areas **300** that are large in size and/or have complex or irregular shapes. According to the present disclosure, the central controller **204** associated with the system **200** transmits the information related to the smaller virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** that have been previously worked on by the earthmoving machine **102** to the second controller **208**.

(69) Further, the system **200** and the method **900** may allow autonomous operation of the compactor **108** once the earthmoving machine **102** has exited the virtual fence (such as, the virtual fence **810**) around the corresponding virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. Specifically, the central controller **204** may generate and transmit the information

pertaining to each virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** to the second controller **208** as and when the earthmoving machine **102** exits the corresponding virtual work area **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. Thus, the compactor **108** may operate on the virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** that may have a smaller area and a definite shape. Accordingly, the virtual segmentation of the work area **300** may improve the autonomous operation of the compactor **108**. The system **200** and the method **900** described herein may also increase a productivity and an efficiency at the worksite **100**.

(70) Further, the central controller **204** generates the optimal direction of movement for the compactor **108** based on the movement of the earthmoving machine **102** in each of the corresponding virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. The optimal direction of movement for the compactor **108** being determined by the central controller **204** may be such that the compactor **108** may take least number of turns, or passes, for effectively performing the compaction operation, on the virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. Additionally, the optimal direction of movement being determined by the central controller **204** may be such that the compactor **108** performs the compaction operation at each desired location of the virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H**. Further, in an example, the optimal direction of movement may be generated along long and continuous edges of the virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** which may allow the compactor **108** to move along straight paths without frequent changes to a steering and/or articulation angle of the compactor **108**, thereby facilitating an increase in productivity of the compactor **108**.

(71) Moreover, the system **200** and the method **900** described herein may perform a vectorized rectangularization of the work area **300**. Further, the virtual segmentation of the work area **300** into the smaller virtual work areas **306A**, **306B**, **306C**, **306D**, **306E**, **306F**, **306G**, **306H** may also ensure that the second controller **208** associated with the compactor **108** receives data corresponding to each location of the work area **300** on which the earthmoving machine **102** may have performed the one or more operations. Thus, the second controller **208** may have exhaustive information of each portion of the work area **300** and therefore, enable autonomous operation of the compactor **108**. In such examples, a likelihood of incomplete compaction operations on the work area **300** that were previously worked on by the earthmoving machine **102** may be minimized.

(72) While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed machines, systems, and methods without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A system for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine, the system comprising: a first controller associated with the earthmoving machine, wherein the first controller receives information pertaining to a work area on which the earthmoving machine is required to perform at least one operation; and a central controller coupled in communication with the first controller, wherein the central controller is configured to: receive, from the first controller, the information pertaining to the work area on which the earthmoving machine is required to perform the at least one operation; analyze the work area for virtually segmenting the work area into a plurality of virtual work areas based on a receipt of the information pertaining to the work area; receive, from the first controller, data indicative of a movement of the earthmoving machine through each virtual work area from the plurality of virtual work areas; determine an optimal direction of movement for the compactor on at least one virtual

work area from the plurality of virtual work areas based on the data indicative of the movement of the earthmoving machine, such that the compactor is configured to move along the optimal direction of movement during the autonomous operation of the compactor, the optimal direction being generated by taking into consideration the movement patterns of the earthmoving machine, and by taking into consideration the geometry of the work area including determining an orientation of passes of the earthmoving machine; generate a virtual fence around the at least one virtual work area from the plurality of virtual work areas based on generation of the optimal direction of movement for the compactor; transmit information pertaining to the optimal direction of movement to the compactor when the earthmoving machine is outside the virtual fence generated around the at least one virtual work area; and control the compactor to move in the optimal direction.

2. The system of claim 1, wherein the central controller is configured to: receive, from the first controller, a plurality of position inputs associated with the earthmoving machine, wherein each position input from the plurality of position inputs is indicative of the position of the earthmoving machine on the at least one work area with respect to time; generate a plurality of vectors based on an analysis of the plurality of position inputs; and determine the optimal direction of movement for the compactor based on an average dot product of the plurality of vectors.

3. The system of claim 1, wherein the central controller is configured to analyze at least one of a plurality of convex vertices and at least one of a plurality of concave vertices on the work area for virtually segmenting the work area into the plurality of virtual work areas.

4. The system of claim 3, wherein the central controller is configured to virtually segment the work area into the plurality of virtual work areas based on an angle defined by the plurality of concave vertices, and wherein the plurality of virtual work areas includes at least one of a structured virtual work area and an unstructured virtual work area.

5. The system of claim 3, wherein the central controller is configured to determine at least one of an entry point for the compactor and an exit point for the compactor based on an analysis of the plurality of convex vertices.

6. The system of claim 1, wherein an area of the virtual fence is configured to be greater than an area of the at least one virtual work area.

7. The system of claim 6, wherein the central controller is configured to: determine the optimal direction of movement based on a conclusion of the at least one operation by the earthmoving machine; and transmit, to a second controller associated with the compactor, the information pertaining to the optimal direction of movement when the earthmoving machine is outside the virtual fence, wherein the second controller is coupled in communication with the central controller.

8. A method for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine, the method comprising: receiving, by a first controller associated with the earthmoving machine, information pertaining to a work area on which the earthmoving machine is required to perform at least one operation; receiving, by a central controller, information pertaining to the work area on which the earthmoving machine is required to perform the at least one operation from the first controller, the central controller is coupled in communication with the first controller; analyzing, by the central controller, the work area for virtually segmenting the work area into a plurality of virtual work areas based on a receipt of the information pertaining to the work area; receiving, by the central controller, data indicative of a movement of the earthmoving machine through each of the plurality of virtual work areas from the first controller; determining, by the central controller, an optimal direction of movement for the compactor on at least one virtual work area from the plurality of virtual work areas based on the data indicative of the movement of the earthmoving machine, such that the compactor is configured to move along the optimal direction of movement during the autonomous operation of the compactor, the optimal direction being generated by taking into consideration the movement patterns of the earthmoving machine,

and by taking into consideration the geometry of the work area including determining an orientation of passes of the earthmoving machine; generate, by the central controller, a virtual fence around the at least one virtual work area from the plurality of virtual work areas based on generation of the optimal direction of movement for the compactor; transmit, by the central controller, information pertaining to the optimal direction of movement to the compactor when the earthmoving machine is outside the virtual fence generated around the at least one virtual work area; and control, by the central controller, the compactor to move in the optimal direction.

9. The method of claim 8 further comprising: receiving, from the first controller, a plurality of position inputs associated with the earthmoving machine, wherein each position input from the plurality of position inputs is indicative of the position of the earthmoving machine on the at least one work area with respect to time; generating, by the central controller, a plurality of vectors based on an analysis of the plurality of position inputs; and determining, by the central controller, the optimal direction of movement for the compactor based on an average dot product of the plurality of vectors.

10. The method of claim 8 further comprising analyzing, by the central controller, at least one of a plurality of convex vertices and at least one of a plurality of concave vertices on the work area for virtually segmenting the work area into the plurality of virtual work areas.

11. The method of claim 10 further comprising virtually segmenting, by the central controller, the work area into the plurality of virtual work areas based on an angle defined by the plurality of concave vertices, wherein the plurality of virtual work areas includes at least one of a structured virtual work area and an unstructured virtual work area.

12. The method of claim 10 further comprising determining, by the central controller, at least one of an entry point for the compactor and an exit point for the compactor based on an analysis of the plurality of convex vertices.

13. The method of claim 8 further comprising generating, by the central controller, the virtual fence such that an area of the virtual fence is configured to be greater than an area of the at least one virtual work area.

14. The method of claim 13 further comprising: determining, by the central controller, the optimal direction of movement based on a conclusion of the at least one operation by the earthmoving machine; and transmitting, by the central controller, the information pertaining to the optimal direction of movement to a second controller associated with the compactor when the earthmoving machine is outside the virtual fence, wherein the second controller is coupled in communication with the central controller.

15. A non-transitory computer readable medium having computer executable instructions for performing a method for generating a work plan for autonomous operation of a compactor in tandem with an earthmoving machine, the method comprising: receiving information pertaining to a work area on which the earthmoving machine is required to perform at least one operation; analyzing the work area for virtually segmenting the work area into a plurality of virtual work areas based on a receipt of the information pertaining to the work area; receiving data indicative of a movement of the earthmoving machine through each of the plurality of virtual work areas; determining an optimal direction of movement for the compactor on at least one virtual work area from the plurality of virtual work areas based on the data indicative of the movement of the earthmoving machine, such that the compactor is configured to move along the optimal direction of movement during the autonomous operation of the compactor, the optimal direction being generated by taking into consideration the movement patterns of the earthmoving machine, and by taking into consideration the geometry of the work area including determining an orientation of passes of the earthmoving machine; generating a virtual fence around the at least one virtual work area from the plurality of virtual work areas based on generation of the optimal direction of movement for the compactor; transmitting information pertaining to the optimal direction of movement to the compactor when the earthmoving machine is outside the virtual fence generated

around the at least one virtual work area; and controlling the compactor to move in the optimal direction.

16. The non-transitory computer readable medium of claim 15 further comprising: receiving a plurality of position inputs associated with the earthmoving machine, wherein each position input from the plurality of position inputs is indicative of the position of the earthmoving machine on the at least one work area with respect to time; generating a plurality of vectors based on an analysis of the plurality of position inputs; and determining the optimal direction of movement for the compactor based on an average dot product of the plurality of vectors.

17. The non-transitory computer readable medium of claim 15 further comprising analyzing at least one of a plurality of convex vertices and at least one of a plurality of concave vertices on the work area for virtually segmenting the work area into the plurality of virtual work areas.

18. The non-transitory computer readable medium of claim 17 further comprising virtually segmenting the work area into the plurality of virtual work areas based on an angle defined by the plurality of concave vertices, wherein the plurality of virtual work areas includes at least one of a structured virtual work area and an unstructured virtual work area.

19. The non-transitory computer readable medium of claim 17 further comprising determining at least one of an entry point for the compactor and an exit point for the compactor based on an analysis of the plurality of convex vertices.

20. The non-transitory computer readable medium of claim 15 further comprising generating the virtual fence such that an area of the virtual fence is configured to be greater than an area of the at least one virtual work area.
